

Numerical Methods

BCA — Semester 3 — — Answer Sheet

Subject Code:

Note: This answer sheet is currently a template. Detailed answers will be added progressively. Students are encouraged to refer to textbooks and class notes.

Group B

1. Define true error and relative error. Derive the bisection method for solving non-linear equation and using this method solve $(2x^3 - 2x - 5)$ with initial $(x_0 = 1)$ and $(x_1 = 2)$. Calculate upto $(10^{\{th\}})$ iteration.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

2. What are the applications of interpolation? Differentiate between interpolation and regression. Consider the following data points estimate the $(f(10))$ using Lagrange's interpolation.
x56911y13141516

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

3. What do you mean by numerical integration? Write any one application of numerical integration. Write an algorithm and c program to implement multi-segment trapezoidal rule.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

4. Construct the divided difference table for the following data value and find first and second order derivative at $(x = 2)$.
x124810y0152127

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

5. Why it is better to use composite Simpson's $(\frac{3}{8})$ rule instead of Simpson's $(\frac{3}{8})$ rule? Find the value of integration for following data set using Simpson's $(\frac{3}{8})$ rule.
x0123456y10.50.20.10.05880.03850.027

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

6. What is least squares method of fitting a function? Fit the second order polynomial for the following data values.
x1234567y2678101215

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

7. Solve the following system of linear equations using Gaussian elimination method. $(2x + 2y + z = 12)$ $(3x + 2y + 2z = 8)$ $(5x + 10y - 8z = 10)$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

8. Solve the following system of linear equations using Gauss-Seidal method. $(10x + y + z = 12)$ $(2x + 10y + z = 13)$ $(2x + 2y + 10z = 14)$

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

9. Define eigen value and eigen vector. Explain how shooting method is used to solve boundary value problem.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

10. Find the approximate value of (y) when $(x = 0.6)$ of $(\frac{dy}{dx} = 1 - 2xy)$, given that $(y = 0)$ when $(x = 0)$ with $(h = 0.2)$ using Heun's method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

11. Consider a steel plate of size $24\text{cm} \times 24\text{cm}$. If two of the opposite sides are held at 100°C and the other two opposite sides at 0°C , find the steady state temperatures of interior points, assuming a grid size of $8\text{cm} \times 8\text{cm}$.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

12. Write an algorithm for Horner's method. Evaluate the polynomial $(f(x) = x^4 + 3x^3 + 5x^2 + 7x + 9)$ at $(x = 2)$ by using Horner's method.

Answer: [Detailed answer will be added soon. Students should write a comprehensive answer covering key concepts, examples, and diagrams where applicable.]

These answer sheets are prepared for educational purposes. Students are encouraged to verify with official textbooks and course materials.